

HOSPITAL AI PROJECT

Deep Research Dossier

Diagnostics Interpretation & Hospital Operations — Directional Research, Full Depth

Anchors	(1) ~1,500-patient real dataset (2) Diagnostics: senior-to-junior-doctor result interpretation, PONV as one instance (3) Operations/logistics
Method	7 deep research passes, live website exploration, not literature summary
Status	Research and scoping only — nothing built yet
Date	22 August 2026

WHY THIS DOCUMENT IS LONG

This is a directional research dossier, not a summary. Every company, number, and source below was independently verified through live exploration — real product pages, published safety cases, peer-reviewed studies, regulatory PDFs — not compressed into takeaways. It is organized around the project's two real directions: **diagnostics interpretation** (explaining results simply, senior-level judgment made usable by a junior doctor) and **hospital operations/logistics**, with the patient dataset and the trust/governance layer as supporting threads underneath both.

Part 1 — The Dataset, Pushed as Far as It Goes

Everything downstream starts from the real ~1,500-patient PONV dataset. Before looking outward at the industry, the dataset itself was pushed past descriptive statistics into formal, defensible statistics.

1.1 From percentages to real statistics

The original pass produced descriptive findings (who had PONV, by what %). This was insufficient — percentages don't control for confounding, so a second pass ran proper multivariate logistic regression with odds ratios and confidence intervals.

Predictor	Univariate OR	Multivariate aOR	95% CI	Verdict
Motion sickness history	2.99	2.91	2.11–4.00	Independent, strong
Previous PONV	2.35	2.52	1.70–3.75	Independent, strong
Female sex	2.27	2.10	1.58–2.80	Independent, moderate
Bellville score	—	excluded	—	Data leakage — see 1.2
Ondansetron given	0.4	0.51	0.29–0.89	Confounded by indication — see 1.3

Model performance: logistic regression AUC 0.677–0.691 depending on feature set. This is **in line with, not below, published PONV literature** — confirmed directly against the diagnostics-automation research in Part 3: published AI PONV models beat the free Apfel score by roughly 4 percentage points (68.6% vs 64.3% accuracy), and Apfel itself tops out around AUC 0.7 in most validation cohorts. Our numbers are not an underperforming model — they are what an honest PONV model looks like at this patient volume.

1.2 The Bellville score problem — solved, one question remains

A specific question raised by the raw data: what does a Bellville score of 10 or 11 mean, when the field is scored on what appears to be a 1–10 range in this dataset? Investigated directly. Finding: **no such scale exists in the clinical literature**. The real, published Bellville nausea/vomiting score is 0–3 (0 = none, 1 = nausea only, 2 = vomiting/retching, 3 = both). Values of 10–11 appearing in the dataset cannot be a single Bellville reading.

Two live possibilities, not yet resolved: (a) data-entry error, or (b) the field is actually a summed 24–48-hour cumulative score across multiple assessment points, which some hospital charting systems do use informally under the same field name. **This needs direct clinical confirmation** — it is not resolvable from the dataset alone.

Regardless of the answer, the score was correctly excluded from the risk model: if Bellville is measured *after* the PONV event occurs, including it as a “predictor” is data leakage — it would make the model circular (predicting the outcome using a measurement of the outcome).

1.3 Ondansetron — confounding by indication, verified against real published data

The raw data shows patients **given** ondansetron had **lower** PONV rates in some cuts and the opposite pattern in others — counter-intuitive on its face. Deep research resolved this: it is a textbook, well-published pattern

called **confounding by indication** — clinicians preferentially give anti-emetics to patients they already judge to be high-risk (prior PONV history, motion sickness, female sex, laparoscopic surgery), so the drug's association with outcome is contaminated by why it was prescribed in the first place.

Cross-validated directly against a real 76,703-patient published study with the identical pattern documented — this is not a data quality problem in our dataset, it is expected, reproducible clinical epidemiology. The correct handling is to model prescribing *propensity* separately from outcome, or treat ondansetron as a confounder to adjust for rather than a predictor to trust at face value. This is a real methodological finding worth stating plainly: the dataset behaved exactly as real clinical literature predicts it should.

1.4 Cross-validation against independent India-specific research

Our own odds ratios were checked against independently published India-specific PONV studies rather than taken on faith:

Predictor	Our aOR	Independent published OR	Agreement
Motion sickness	2.91	2.5	Close — same direction, similar magnitude
Previous PONV	2.52	3.2	Same direction, plausible spread

This convergence matters for credibility: the dataset is not an outlier or a data-quality artifact — it reproduces known clinical epidemiology at a small-hospital scale.

1.5 What is usable, what is missing

Category	Status
Usable, statistically validated predictors	Motion sickness, previous PONV, sex, surgery type, ASA class
Excluded (data leakage)	Bellville score as a predictor
Present but requires careful handling	Ondansetron (confounded by indication)
Missing from dataset entirely	Intraoperative data (anaesthetic duration, blood loss, exact opioid dosing) — known limiters of AUC ceiling in the wider literature too

Part 2 — Diagnostics: Senior Judgment, Made Usable by a Junior Doctor

This is the project's primary direction: not “AI replaces the doctor,” but **senior-level interpretation compressed into a form a junior doctor can act on quickly and safely**. PONV risk prediction is one instance of this pattern, not the whole of it. Two research passes went directly at this — one on live clinical-decision-support products, one specifically at the junior-doctor interpretation workflow.

2.1 The industry-wide pattern: automate the confident cases, not the hard ones

Across ~40 real companies examined directly (not literature reviews of them — product pages, published validation papers, regulatory filings), one pattern recurs without exception: **nobody automates the ambiguous, high-stakes decision**. They automate clearing out the confident, low-risk cases so a human's attention concentrates on the genuinely uncertain ones.

- **Oxipit / ChestLink** — the sharpest real example. Auto-clears chest X-rays only when the model is 99.9%+ confident they are completely normal. Everything else, including all uncertain cases, goes to a radiologist untouched. Runs a three-phase staged rollout (shadow mode → partial autonomy → full autonomy) with human sign-off gating every phase transition.
- **CLSI AUTO-10** — the actual industry standard for lab-result autoverification. Rule-based, not ML: a result only auto-releases if it falls inside a pre-validated “normal enough” band; anything outside routes to a human. This is the direct, boring, well-proven precedent for a “confident-normal auto-clear, everything else to a human” design.
- **Rad AI** — measures success not by accuracy claims but by **edit rate**: how often a radiologist accepted the AI's draft with zero or one correction (87–98% of the time in their real deployment data), plus a separate **error-catch rate** (the AI catches a mistake the human made, ~5% of the time). This is a clean, auditable, non-black-box way to demonstrate the tool helps — directly transferable to a junior-doctor interpretation tool: track how often the junior doctor's action matches or improves on what the tool suggested.
- **Cognita / Mosaic** — same edit-rate philosophy applied to a different diagnostic domain, reinforcing this as an emerging industry-standard evaluation metric, not a one-off.

Sources: Oxipit product/clinical documentation; CLSI AUTO-10 standard; Rad AI and Cognita published deployment metrics; AMA CPT Appendix S taxonomy.

2.2 The official classification this kind of tool falls under

The American Medical Association's CPT Appendix S sorts medical AI into three billing/regulatory tiers, and this single framework resolves a lot of scoping ambiguity in writing:

Tier	What it means	Where PONV/interpretation tool sits
Assistive	Shows data; human interprets everything	—
Augmentative	Calculates a score/risk number; human decides what to do with it	HERE — confirmed fit
Autonomous	AI makes the actual clinical decision	— explicitly out of scope, and per Part 5, against ICMR ethical guidance in India without a human in the loop

A separate, more granular L0–L5 clinical-AI-autonomy framework exists in parallel literature and places the same kind of tool consistently in the low-autonomy band — second independent confirmation of the same scoping conclusion.

2.3 Junior-doctor-specific interpretation design — what actually works

This is the sharpest match to the project's stated direction. Findings specific to translating senior-level judgment into a fast, safe junior-doctor-facing interpretation:

- **Gauge visualization beat every alternative tested** in a rigorous 3-iteration, 25-clinician co-design study on a hospital dashboard (traffic-light gauge vs green/orange/red vs binary green/red). Clinicians explicitly cited “common stoplight understanding.” Full detail and caveats in Part 4.
- **Segments should not be equal thirds** — the same study found clinicians preferred a *dominant yellow zone*, smaller green and red, because the desired mental frame was “stay in the safe zone,” not “am I in the danger zone.” Directly transferable to a PONV risk gauge shown to a junior doctor.
- **No hard-stop interrupting alerts** — clinicians in the same study explicitly rejected mandatory-interrupt alerts in favor of discretionary access. This is independently reinforced by hard incident data in Part 4 (101 deaths tied to alarm-fatigue incidents in one dataset) — two unrelated sources converging on the same design rule.
- **Confidence/uncertainty display is still an open research question, not a solved pattern** — a live UCSF randomized trial (NCT07457840) is currently testing whether showing confidence intervals actually improves diagnostic accuracy, and has not yet reported. One documented risk: confidence intervals can raise a clinician's *trust* in a prediction without raising its *correctness* — the same false-reassurance failure mode already flagged for AI-generated explanations generally. Treat uncertainty display as something to design carefully and test, not something to assume is automatically good UX.
- **Progressive disclosure, conditional type**, is the right form pattern for a PONV-style risk-factor intake form — only show follow-up fields when a prior answer makes them relevant (matches how the Apfel score's four inputs naturally branch).

Sources: PMC9179234 (25-clinician co-design study); NCT07457840 (UCSF trial, ongoing); arXiv 2507.12721 (12-pattern healthcare human-AI interface taxonomy, 12 clinicians + 14 UI designers).

2.4 The two cautionary tales every design decision must be checked against

CAUTION — Epic Sepsis Model

Deployed at hundreds of real hospitals. Vendor-reported performance did not hold up: externally validated AUC **0.63** at Michigan Medicine, **missed two-thirds of actual sepsis cases**, and at Epic's own recommended alert threshold caught only 7% of cases clinicians hadn't already caught themselves — near-zero marginal value. **109 alerts investigated per one true case** — quantified alert fatigue. Epic's own response: hospitals must retrain the model on their own local patient population before deployment. Direct implication: any model built here is valid for the specific source hospital's population only, and that must be stated explicitly, not implied as general-purpose.

CAUTION — Epic Sepsis Model

(continued) Report calibration alongside AUC, not AUC alone — the model's calibration failure was as damning as its discrimination failure, and calibration is the metric most likely to be silently skipped.

CAUTION — Edwards HPI (Hypotension Prediction Index)

A comparably sophisticated intraoperative prediction tool (23 inputs, predicting dangerous blood-pressure drops during surgery) was found in independent validation to perform **no better than watching one basic vital sign directly**. Direct implication: complexity is not evidence of value. Every version of our model must be benchmarked head-to-head against the free, already-in-use Apfel score, and we must be honest if the margin is small — published literature puts the realistic AI-over-Apfel improvement at roughly 4 percentage points, not a breakthrough.

2.5 India-specific deployment reality — a real precedent, not theory

A live, deployed India example was found directly on point: a TB-screening AI tool (qXR) rolled out across real Indian hospitals. Documented real barriers, not hypothetical ones:

- Hospitals refused to grant the software administrator access to their computers
- Some X-ray machines saved files in a format the AI tool could not read
- Local doctors initially resisted the tool simply because they were never consulted or trained on it before rollout

These are the practical, human-side failure modes to plan for now, not technical ones — directly reinforced in Part 5's "pilotitis" finding: *"The failure of AI pilots to scale is not a technical failure but a systems failure... frontline staff were never consulted, trained, or given a reason to care."*

Sources: qXR India deployment case study; ICMR March 2023 Ethical Guidelines for AI in Biomedical Research/Healthcare; India national "Smart Doctor" CDSS rollout (confirmed rule-based, not ML).

Part 3 — Operations & Logistics

The second anchored direction: hospital bed/staff/resource tracking. This module is architecturally and regulatorily distinct from the diagnostics module (see Part 5.1), and its design language is different too — it is monitoring a system over time, not scoring an individual patient.

3.1 The single most load-bearing finding: RAG tiles are the wrong chart type

NHS England ran a national program (“Making Data Count”) training 61 hospital trust boards between Nov 2017 and Jul 2019 specifically to move them **away from** red/amber/green status tiles and **onto** Statistical Process Control (SPC) charts for exactly this kind of operational monitoring. NHS England has since switched its own national performance monitoring from RAG to SPC. The reasoning: RAG tiles on a single snapshot number cause staff to react to ordinary day-to-day noise as if it were a real signal — chasing variation that isn't actually meaningful.

This is directly actionable and free: **NHSRplotthedots** is an open-source R package implementing the exact SPC chart rules NHS trusts now use (which variation counts as a real signal vs. ordinary noise). Even without using R, the underlying rule logic is documented and portable to any stack.

Design conclusion for our ops module: bed occupancy, staffing levels, and resource tracking should be SPC time-series charts, not RAG status tiles. This is a 6-years-of-real-institutional-experience-backed choice, not a stylistic preference.

Sources: NHS England “Making Data Count” program documentation; NHSRplotthedots (GitHub, nhs-r-community).

3.2 Reliability over real-time correctness — the offline/staleness pattern

Deep research on offline-first hospital connectivity converged on one clear rule, directly applicable to a bed-tracking dashboard that may run on unreliable hospital wifi: **a confidently wrong number is worse than an honestly stale one.**

- Form/data submissions should queue locally (not attempt direct real-time writes) and only clear from the queue after a **confirmed** successful upload — so a failed sync retries instead of silently vanishing.
- **For the ops module specifically: no bidirectional sync.** Two staff members updating the same bed record offline creates a real conflict with no automatic correct answer. The safer design is read-only cached data with an explicit, visible “last updated N minutes ago” staleness label — a bed board that admits it might be out of date is safer than one that looks live but isn't.

Sources: Offline-first PWA architecture research (IndexedDB queue pattern, Background Sync API); cross-validated against the Part 4 finding on silent-failure incidents in real EHR systems.

3.3 What automation actually looks like in hospital operations today

Cross-referenced against the diagnostics-automation research in Part 2 — the same “automate the confident, escalate the uncertain” philosophy applies to operations, not just diagnostics. The practical translation for bed/resource tracking: automate routine status updates and flagging of clearly-idle or clearly-overloaded resources; keep allocation *decisions* (which patient gets which bed, which staff member gets reassigned) as human calls, informed by the dashboard rather than made by it — matching the Augmentative tier classification from Part 2.2, applied to logistics rather than diagnosis.

3.4 The realistic bar in an Indian hospital context

Restated from earlier research and reinforced by this pass: the honest baseline this module competes against at most Indian hospitals, including larger ones, is *phone calls and printed/PDF handoffs between shifts*. A working, honestly-labeled, SPC-charted, offline-tolerant dashboard clears that bar comfortably — it does not need to compete with a mature US hospital's real-time IoT-integrated bed-management platform to be a genuine improvement.

3.5 Named vendors, independently verified — not taken from search-engine summaries

Two batches of named logistics/diagnostics-automation vendors were surfaced during this research and independently re-verified directly against each vendor's own website, rather than trusted from the search-engine summaries that first surfaced them. This distinction matters: several claims in the original summaries did not hold up, and are flagged below as debunked rather than silently dropped.

Batch 1 — India-relevant logistics/LIMS vendors

Vendor	India presence	Verified strongest claim	Unverified / debunked claim
CrelioHealth	Yes — 2 named Indian labs (Zorg, Insta)	70+ analyzer integrations, real product	OCR requisition digitization, auto-reorder inventory — NOT found on their own site
SmartCarePlus	India-native	100+ analyzers via ASTM/HL7; WhatsApp report delivery; public INR pricing	No deployment-scale numbers published
nuVizz (RoboDispatch)	None found	Real AI dispatch algorithm (proximity/capacity/workload matching)	No India footprint at all
Swisslog Healthcare	None found	OpenLog tube-to-analyzer delivery; PillPick unit-dose sachet packing	No throughput or error-rate numbers published
CXT Software	None found	Barcode-enforced chain-of-custody — CONFIRMED	“Photo proof” specifically unconfirmed
LogiNext	HQ in US despite India origin; no India clients named	IoT cold-chain monitoring + AI routing, concrete metrics	“300+ healthcare providers” asserted with zero names
Dematic / Armstrong Dematic	Yes (Armstrong Dematic, India partner)	General ASRS/robotics capability	Zero healthcare or hospital deployment evidence found anywhere
Onymos / DockKnow	None (not India-relevant)	OCR intake: 84% error reduction, 99.7% validation accuracy — most technically detailed intake system found	Architecture pattern only, not a deployable India vendor

Finding, stated plainly: no single vendor is an India-proven template for this project's ops/diagnostics stack. The one vendor combining India-native operation with a specific, checkable technical claim is SmartCarePlus. The right architectural response is componentized pattern-borrowing — OCR intake in the style of DockKnow, dispatch logic in the style of RoboDispatch/LogiNext, workflow enforcement in the style of CXT — built ourselves, not licensed wholesale from one vendor.

Batch 2 — Drone logistics & industrial biobanking: verified, then excluded on scope grounds

A second batch of vendor names (Matternet, Zipline, Omnicell, LiCONiC, Hamilton Storage) was independently verified against primary sources. All are real technologies with genuine specifications — but every one describes deployments built for national blood-supply networks or large-scale research biobanks, not a single hospital department. They are recorded here for completeness and explicitly excluded from the implementable-vendor list.

Vendor	Verified fact	Correction to original claim	Scope verdict
Matternet	M2 drone: 2kg payload, 20km range, automated battery-swap stations — confirmed exact match to primary source	None — claim was accurate	Excluded — built for inter-facility/inter-campus corridors, not one building
Zipline	P1 platform: fixed-wing, parachute delivery, ~96km radius	The pasted spec describes Zipline's deprecated P1 platform, not their current P2 (no parachute, ~16–39km, hover-and-lower tether)	Excluded — national health-ministry / regional hospital-system scale
Omnicell XR2	Real, current product name; barcode-scanning safeguard confirmed	“Near-0% error rate” is not on Omnicell's own page — unconfirmed, do not cite	Category-relevant but enterprise-scale hardware, not a project-scale reference
Swisslog TransLogic (hemolysis claim)	RFID confirmed for chain-of-custody access control; hemolysis-in-transport is real, peer-reviewed science	“RFID auto-adjusts air pressure to prevent hemolysis” is not a documented Swisslog feature — appears fabricated, conflates three real but separate facts	Category relevant (pneumatic tubes are standard single-hospital infrastructure); cite only the confirmed mechanism
LiCONiC BiOLiX	Product real; robotic sample retrieval confirmed	“–10–196°C” is wrong — the actual line operates –80°C to room temperature, not cryogenic depth	Excluded — industrial research-biobank scale
Hamilton Storage Verso	Capacity confirmed (100,000+ tubes depending on config); throughput 1,500 tubes/hr confirmed	“Retrieval under 20 seconds” — not found in any source, unconfirmed	Excluded — drug-discovery/large-biobank infrastructure

Three lab-analyzer specs from the same batch (Siemens Atellica, Abbott Alinity, Roche Cobas Pro) were also verified and, unlike the items above, **are correctly scaled for a single hospital's core lab** — confirmed: Atellica's chemistry module runs ~1,800 tests/hour via patented bi-directional Magline transport (10x faster than conventional conveyors); Alinity's vertical stacked-module design delivers Abbott's own newsroom-stated ~50% core-lab space saving; Cobas Pro's “on-the-fly” continuous reagent loading supports 24/7 operation with ~5 minutes/day operator intervention. These are legitimate to cite as the kind of hardware that generates the lab data a junior-doctor interpretation tool would consume.

The FHIR question — resolved, not left open

A direct conflict arose between this dossier's existing architecture recommendation (Part 4.5: skip full FHIR, use file-based model registry + REST + FHIR-style field names as a vocabulary hedge only) and a pasted claim

recommending a blanket “shift to FHIR.” This was investigated directly rather than left as an open disagreement.

Verdict: the existing recommendation is correct and is being kept as-is. The “FHIR is mandatory” framing traces back to a real but narrow US rule — ONC certification for commercial EHR products seeking Meaningful-Use incentive status — which does not apply to a single-department, student-built internal tool with no external data-exchange partner. Even FHIR-forward industry sources draw the identical scale line this dossier already drew: “for everyone else [without dedicated IT departments and million-dollar budgets], [FHIR] can be overkill and unaffordable,” and industry consensus at the scale where FHIR is justified is *coexistence* with HL7v2 for backend operations, not wholesale replacement. This dossier's approach — FHIR-style field naming and LOINC/SNOMED vocabulary, without importing FHIR's resource-graph, versioning, or server infrastructure — already captures the one real benefit available at this project's scale.

Part 4 — Trust & Governance: Making Both Directions Safe to Build

This thread exists to serve Parts 2 and 3, not as a separate direction. It covers, from a developer's seat, the UI, frontend/backend architecture, database schema, and error-prevention/legal layer needed to build either module responsibly.

4.1 The regulatory headline that changes scoping today

NEW — CDSCO finalized Medical Device Software guidance, 21 July 2026

Document CDSCO/MD/GD/MDSW/01/2026, issued one month before this research. Supersedes the earlier Oct 2025 draft. The critical sentence: exclusions are **“function-specific, not product-specific.”** This means the diagnostics module and the ops module are **not one product for regulatory purposes**, even inside the same codebase — ops/bed-tracking is out of scope, PONV/result-interpretation is in scope, and if they share a deployment, the regulated surface contaminates the whole system. **Architecturally separate them from day one.** The guidance also formalizes an Algorithm Change Protocol (predefined bounded retraining rules, designed in upfront) and explicitly names “bias, model drift and hallucination” as post-market monitoring obligations, referencing IEC 62304, ISO 14971, IEC 82304-1.

Sources: CDSCO guidance PDF (cdsco.gov.in); National Law Review analysis; Emergo by UL.

4.2 Risk-score visualization — full detail

(Referenced in Part 2.3; full findings [here](#).) The 3-iteration, 25-clinician co-design study on a hospital dashboard is the strongest empirical source found in this entire research effort:

- Gauge with green/yellow/red beat both green/orange/red and binary green/red — “common stoplight understanding” cited directly by participants.
- Segments are NOT equal thirds — final version used smaller green and red zones with a dominant yellow, reframing the goal as “stay in the safe zone” rather than “avoid the danger zone.”
- The predicted number sits on the *exterior* of the gauge, not inside it — a small, specifically tested detail.
- Final design paired the overall gauge with per-condition sub-gauges, a 3-day directional trend indicator, and a plain-text box of patient context.
- Clinicians explicitly rejected hard-stop mandatory alerts, wanting discretionary access instead.

Two honest caveats the source study itself raises: one participant was colourblind and could not properly see the yellow zone — the team shipped traffic-light colours anyway because the majority preferred them, a documented accessibility failure, not a hypothetical one. Second, the study states plainly: “there is a lack of validated thresholds for the probability of an [event]” — they had no principled basis for exactly where green becomes yellow. **We will have the identical problem with PONV risk cutoffs and should say so rather than pretend the threshold is more rigorous than it is.**

4.3 Colourblind-safe palette — the concrete fix for the accessibility gap above

- Blue is the safe anchor colour — protanopia and deuteranopia (the common ~8%-of-males deficiencies) barely affect blue perception. Safe pairs: blue/orange, blue/red, blue/brown.

- **Practical resolution: replace the red/green axis with blue/orange**, keep the luminance ramp monotonic so it survives greyscale printing, and **always dual-encode** — colour plus numeral plus position plus icon shape, never colour alone. This directly fixes the colourblind participant's failure documented in 4.2.
- WCAG minimums still apply: 4.5:1 contrast for normal text (AA), 7:1 (AAA). Verification tools: ColorBrewer (CVD-safe colormaps), Adobe Color accessibility checker.

4.4 Frontend & offline architecture

A real, directly comparable precedent exists: the **RCPCH Digital Growth Charts** — an actual NHS-facing clinical tool, published safety case included, built by a small team for a single clinical purpose — ships React for its charting component. Reasoning that transfers directly to our project: accessibility-library maturity (React Aria/Radix/Headless UI give WCAG-correct form components out of the box, which matters for a form-heavy clinical intake tool), and successor-hiring depth for handover (directly relevant given the sustainability risk in 4.7).

Offline pattern (detailed in Part 3.2, restated for the diagnostics module specifically): the PONV/result-interpretation module should be **offline-capable capture with a one-way outbound queue** — never bidirectional clinical-record sync. The universal warning found across sources: *“Sync is where everything breaks in offline-first systems, and users often never see it happen.”* A silently failed sync in a clinical tool is a patient-safety hazard, not a UX annoyance.

4.5 Backend, schema, and model versioning

Decision	Choice	Why
API style	REST, not GraphQL	The URL path is itself the audit-log record (e.g. GET /patients/{id}/ponv-assessment). GraphQL's single POST /graphql endpoint requires parsing the query to know what was accessed — solved at enterprise scale, not at student-team scale.
FHIR	Skip full implementation; use FHIR-style field names + LOINC/SNOMED codes where they exist	Full FHIR is an interoperability spec, not free architecture — version migrations are documented as painful and a generic model fits specialized data (like PONV factors) awkwardly. The hedge costs almost nothing and preserves future integration optionality.
Model versioning	File-based registry: immutable artifact + content hash + a model_versions table (version, hash, training snapshot, metrics, approver, activation date)	MLflow beyond local use needs a tracking server + DB + object store — real DevOps burden a student team shouldn't take on. Every prediction row must store its model_version_id — non-negotiable, required for any future drift audit or the CDSCO Algorithm Change Protocol.
Audit log	Append-only; UPDATE/DELETE revoked at the database role level (not just app code); hash-chained rows for tamper-evidence	Enforced by DB grants, not application trust. Capture actor, patient/record, action, timestamp, IP, device, session ID. DPDP Rule 6 requires 1 year retention, HIPAA-style practice suggests 6 — design retention as a config value, not a hardcoded assumption.

Two-factor patient identity confirmation on every clinical write is a specific, well-evidenced recommendation — wrong-patient record mismatches are documented at up to 20% of patients in some studies, and one real hospital logged over 5,000 wrong-patient orders in a single year. A perfectly accurate PONV score attached to the wrong patient is a harm event, not a near-miss.

4.6 What actually goes wrong in deployed hospital software — real incident data

Real incident: VA / Oracle-Cerner EHR (Mann-Grandstaff)

~150 documented cases of patient harm. Over 11,000 orders for specialty care, labs, and other services **silently disappeared** into an “unknown queue” with no alert to anyone. The single clearest real-world argument for the rule: **every queue in the system needs a visible depth counter and an age alarm**. This is the same underlying bug class as the offline-sync failure warning in 4.4 — two independent sources converging on “silent failure is the top killer,” not “wrong prediction is the top killer.”

Broader incident data, all independently sourced:

- 18,000 logged EHR safety events (2007–2018), 3% causing real patient harm, including seven deaths.
- 170 alarm-management incidents (2014–2018), 101 of them involving patient deaths — directly corroborates the “no hard-stop alerts” rule from Part 2.3 with hard mortality numbers, not just a clinician preference.
- Contributing-factor breakdown across incidents: user error 17%, incorrect information already in the record 16%, **copy/paste and auto-pre-population errors 14%**, paper-to-electronic conversion errors 13%, system/software design 12%. Direct implication: **do not auto-populate or pre-fill clinical fields “to help”** — it is a documented harm mechanism, not a convenience.

4.7 Clinical safety case — a real template that fits a student team

The single most directly reusable artifact found in this whole research effort: the RCPCH Digital Growth Charts project (introduced in 4.4) publishes its complete Clinical Safety Case Report openly. Its structure: Document Controls → Introduction → System Definition → Intended Use (including an explicit disclaimer against unauthorized self-hosting/deployment) → Clinical Risk Management System with a named Clinical Safety Officer → Clinical Risk Analysis → Hazard Log → Test Summary.

Their entire hazard log has **five entries**, each scored severity × likelihood (API unavailability, wrong data entry, wrong measurement units, incorrect calculation, unauthorized self-hosting misuse — all scored “Acceptable” risk with proportionate mitigations: static testing against ~4,000 real records signed off by a named statistician, automated tests, peer code review, input validation ranges, and documented warnings).

This is directly copyable: same document shape, five to eight PONV-specific hazards substituted in. A student team can realistically produce this; it is not a hypothetical enterprise-scale compliance burden.

Underlying standard: IEC 62304 governs software safety classification. Two facts worth stating precisely: Class A (no possible injury) is explicitly described in the standard's own guidance as “*rare for any software claiming a clinical purpose*” — don't assume the tool qualifies for the lightest tier just because it “feels” low-risk. Second, the standard does not mandate waterfall process — it specifies *that* requirements and release documentation exist (§5.2, §5.8 minimum for Class A), not *when* in the development process. An agile build is compliant.

Sources: RCPCH Digital Growth Charts published Clinical Safety Case Report; IEC 62304; Johner Institute and Greenlight Guru guidance.

4.8 Liability — India-specific, and who actually carries it

India has no AI-specific medical negligence statute; disputes fall under existing principles — the Bolam test as modified by Supreme Court precedent, civil liability under the Consumer Protection Act, potential criminal liability under IPC 304A. The consistent finding across sources: **the clinician retains liability** when acting on AI-assisted decision support — if a doctor follows a tool's suggestion without applying independent clinical judgment and it turns out wrong, the doctor bears the risk, not the tool's builders, under the current “learned intermediary” framing (which is being actively challenged as AI systems move toward more autonomy generally, though not for an Augmentative-tier tool like ours).

A second, easily overlooked exposure: the hospital itself can be held liable for failing to exercise due diligence in selecting, operating, or monitoring a deployed tool — meaning the deploying hospital carries real exposure by adopting this, and that is worth surfacing directly and early rather than letting it be discovered later.

Design consequences that follow directly: never present a single recommendation as the only option; log every clinician override of the tool's suggestion with a reason code (this doubles as liability protection **and** the best available drift-detection signal, per Part 4.9); keep intended-use and limitation statements visible in the product UI itself, not buried in a separate document.

Sources: SCC Times on Indian medical device liability; Indian Journal of Health and Medical Law; npj Digital Medicine on prescribing-advice liability.

4.9 Drift monitoring and bias auditing — student-feasible methods

Counter-intuitive finding worth stating precisely: **monitoring model performance alone is not a reliable proxy for detecting data drift** — both input-drift and performance-drift need separate instrumentation. Three drift types: data drift (inputs change), label drift (outcome distribution changes), concept drift (the input→outcome relationship itself changes).

Fully student-feasible, no MLOps platform required: Kolmogorov–Smirnov test for continuous features, Population Stability Index (PSI > 0.25 as the accepted significance threshold), and — conveniently — the same SPC charting technique recommended for the ops dashboard in Part 3.1 doubles as the drift-monitoring visualization, meaning one implementation serves both modules.

For bias auditing, the recommended approach is comparing **probability calibration curves across subgroups** (sex, age band, ward/payer class as a socioeconomic proxy) rather than chasing multiple incompatible fairness metrics at once (demographic parity, equalized odds, and calibration cannot all be satisfied simultaneously — a mathematical fact, not a design failure). The sharper, harder question raised directly by this research: is our outcome label really “*patient had PONV,*” or actually “*PONV was recorded*” — which bakes in recording bias by ward, shift, or staff seniority as if it were a real clinical signal. This is where label-choice bias famously originates in other well-documented AI harm cases, and it's a question worth asking of our own dataset directly.

4.10 Consent — a real, documented gap and a real opportunity

A US national survey (University of Michigan) found patients overwhelmingly expect to be informed and actively involved when AI is used in their care, and that opt-out-style notification is likely insufficient to meet that expectation. Applied to our two modules, the split is clean: the ops module is aggregate, non-patient-facing operational data — general notification suffices; the PONV/diagnostics module directly influences an individual clinical decision — **explicit consent territory**.

Direct search found no Indian hospital that publishes its AI consent policy. This is a genuine, verified gap — and also a real opportunity: drafting a clear consent addendum together with the clinical stakeholder would be a legitimate, original contribution, and under DPDP Act 2023, health data is sensitive data requiring consent that is free, informed, specific, and withdrawable.

This directly reinforces the project's current, correct instinct: PHI has deliberately not been copied into the project folder pending clearance. That should remain the case — the consent/ethics basis needs to be documented *before* any data transfer, not after.

4.11 Sustainability — the highest-probability failure mode, and it isn't technical

No literature addresses “what happens to a student-built hospital tool after graduation” directly — but the identical failure pattern is heavily documented under a different name: **“pilotitis.”** Direct, quotable findings: *“High pilot activity but low policy integration.” “The failure of AI pilots to scale is not a technical failure but a systems failure.” “Once funding ends, servers go offline, trained staff move on, and policymakers shift priorities.”*

For this project, substitute: once the students graduate. This is assessed as the single highest-probability failure mode across the entire research effort — higher than any modeling or technical risk covered in Parts 1–4. Concrete, actually-actionable countermeasures: name a hospital-side owner *before* building anything (not a user — someone with the mandate and credentials to actually run it); default to boring, self-hostable infrastructure (Postgres + a container beats a managed service tied to a student's account); state a fixed supported-until date in the product itself, so an honest sunset happens instead of a zombie tool silently giving stale advice after nobody maintains it; and treat the documentation — the safety case, the hazard log, the schema — as a deliverable in its own right, since it outlives any specific version of the code.

Sources: Frontiers in Digital Health, “From pilot to policy: why AI health interventions fail to scale in developing countries”; qXR India deployment case study (Part 2.5).

Part 5 — Synthesis: What This Means for Scoping

Pulling the four parts above into direct, decidable statements — not a re-summary, a set of positions this research now supports.

5.1 On the two-module split

Confirmed as correct, and now with a regulatory reason attached, not just a logical one: CDSCO's July 2026 guidance treats exclusions as function-specific. The diagnostics module and the ops module must be **architecturally separated** — separate services, ideally separate deployments — or the regulated diagnostics surface contaminates the unregulated ops surface.

5.2 On what the diagnostics module actually is

Confirmed as Augmentative tier (AMA CPT Appendix S) / low-autonomy tier (independent L0–L5 framework): the tool calculates and presents a risk score; the doctor decides. This is not a downgrade from the original ambition — it is the design pattern every real, successful, non-cautionary-tale product in Part 2 actually follows. The senior-to-junior interpretation framing at the core of this direction is best served by the Rad AI edit-rate evaluation pattern (track how often the junior doctor accepts vs. corrects the tool's suggestion) and the gauge/dominant-yellow-zone visualization from Part 4.2.

5.3 On the ops module

SPC charts, not RAG tiles — backed by six years of real NHS institutional experience, not a stylistic guess. Staleness-labeled, one-way, non-bidirectional data sync given hospital connectivity is unreliable.

5.4 On safety and trust

A real, copyable template exists (RCPCH's five-hazard safety case). A student team can realistically produce an equivalent for PONV. The two biggest real-world failure patterns to design against from day one are silent failures (Cerner's 11,000 lost orders) and alert fatigue (101 deaths tied to alarm incidents) — both are design decisions available to us now, not compliance paperwork for later.

5.5 The one finding that matters most outside the code itself

Sustainability, not technical performance, is the highest-probability failure mode. A named hospital-side owner, boring self-hostable infrastructure, and an honest sunset date are cheap to commit to now and expensive to retrofit later. This is the single recommendation from this research pass most worth raising directly and early with project supervision.

STATUS

Nothing has been built. This remains research and scoping only, prepared ahead of the project's requirements-definition stage. Open questions (Bellville score meaning, PHI clearance timing, hospital-side ownership) need direct clinical-stakeholder answers, not further research.

Appendix — Full Statistical Tables

Part 1 presents the headline result of the dataset's risk-factor analysis. This appendix reproduces the complete underlying tables — univariate odds ratios, the multivariate adjustment step in full, and the surgery-type comparison — for readers who want the full statistical trail rather than the summary.

A.1 Univariate odds ratios — each factor considered alone

Before adjustment, several factors that later turn out not to be independent risk factors still appear significant on their own:

Factor	OR	95% CI	p-value
Female gender	2.27	1.72–2.98	<0.0001
Motion sickness history	2.99	2.19–4.08	<0.0001
Previous PONV history	2.35	1.60–3.44	<0.0001
General anaesthesia (vs Spinal/Regional)	1.44	1.11–1.87	0.0058
Age (per year)	0.99	0.98–1.00	0.081 (ns)
BMI (per unit)	0.98	0.95–1.02	0.42 (ns)

A.2 Multivariate (adjusted) odds ratios — the real risk-factor result

Adjusting each factor for all the others in a single model changes the picture meaningfully — this is the step that separates genuine independent risk factors from ones that only appeared to matter because they correlate with the real drivers:

Factor	Adjusted OR	95% CI	Significance
Motion sickness history	2.91	2.11–4.00	p<0.0001 ***
Previous PONV history	2.52	1.70–3.75	p<0.0001 ***
Female gender	2.10	1.58–2.80	p<0.0001 ***
General anaesthesia	1.16	0.88–1.54	p=0.29 (ns)
Age	0.99	0.98–1.01	p=0.36 (ns)
BMI	0.99	0.95–1.02	p=0.45 (ns)

Three factors are genuine, statistically independent risk factors: motion sickness history, previous PONV history, and female gender — each roughly doubling-to-tripling PONV odds, holding the others constant.

The finding worth stating plainly, because it runs counter to clinical intuition: general anaesthesia, age, and BMI all looked like they mattered in the univariate view (A.1), but their apparent effect *disappears entirely* once adjusted for the three real drivers above (A.2). This is the concrete value a formal risk-factor analysis adds over simple percentage breakdowns.

A.3 Surgery type — univariate comparison against the most common procedure

Compared against Lap Chole (laparoscopic cholecystectomy), the single most common procedure in the dataset and used here as the reference category:

Surgery type	OR vs Lap Chole	95% CI	p-value
TURP	0.37	0.16–0.85	0.019
Thyroidectomy	0.47	0.26–0.86	0.015
ORIF Upper Limb	0.48	0.24–0.96	0.039
TKR	0.50	0.30–0.83	0.0075
URSL	0.54	0.32–0.93	0.026
Hernioplasty	0.57	0.35–0.93	0.026
(10 other surgery types)	—	—	not significant vs Lap Chole

Lap Chole carries the highest PONV risk among surgery types with a statistically significant comparison — consistent with published literature, where laparoscopic procedures are a known PONV risk factor. All six procedures listed have significantly *lower* odds than Lap Chole; the ten remaining surgery types in the dataset did not reach statistical significance either way against the Lap Chole reference.

A.4 Model performance and bottom line

Model	AUC	McFadden pseudo-R ²
6 confirmed predictors (multivariate, no surgery type)	0.677	0.067
Same, plus surgery type added	0.691	0.076

Both figures are in-sample (no holdout set — sample size doesn't support a robust train/test split at this scale). This matches published PONV literature almost exactly: models restricted to pre-operative variables consistently land at AUC 0.68–0.72, the same tier as the decades-old Apfel score.

Bottom line: the dataset supports a clean, statistically defensible 3-factor risk model — female gender, motion sickness history, previous PONV history. This overlaps substantially with the published Apfel score's four factors (female sex, non-smoking, PONV/motion sickness history, opioid use); the dataset doesn't capture smoking status or opioid use directly, so a direct one-to-one Apfel comparison isn't possible, but the overlapping factors validate against established literature. Surgery type adds modest additional signal and could be included as a secondary stratification factor. Age, BMI, and anaesthesia type do not hold up as independent risk factors once properly adjusted.